BUS 104: Practice Set — Nonlinear Programming (NLP)

Problems and Solutions

Problem 6-17: Making Change with Minimum Bills and Coins

Formulate and solve an IP model that will decide how to make change for \$22.98, based on the availability of coins and bills shown in the following table:

Denomination	Availability
\$10 bill	1
\$5 bill	4
\$1 bill	1
Quarter (\$0.25)	8
Dime (\$0.10)	1
Nickel (\$0.05)	5
Penny (\$0.01)	6

The objective should be to minimize the total number of bills and coins used to make the change.

Solution to Problem 6-17

1. Decision Variables

Let:

- X_1 = number of \$10 bills used, X_2 = number of \$5 bills used, X_3 = number of \$1 bills used
- X_4 = number of quarters (\$0.25) used, X_5 = number of dimes (\$0.10) used
- X_6 = number of nickels (\$0.05) used, X_7 = number of pennies (\$0.01) used

2. Integer Programming Formulation

$$\text{Minimize Total Bills and Coins} = X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7$$

subject to:

$$10X_1 + 5X_2 + 1X_3 + 0.25X_4 + 0.10X_5 + 0.05X_6 + 0.01X_7 = 22.98 \quad \text{(Amount required)}$$

$$X_1 \le 1,\ X_2 \le 4,\ X_3 \le 1,\ X_4 \le 8,\ X_5 \le 1,\ X_6 \le 5,\ X_7 \le 6 \quad \text{(Availability limits)}$$

$$X_1, X_2, X_3, X_4, X_5, X_6, X_7 \ge 0 \text{ and integer} \quad \text{(Integer decisions)}$$

3. Optimal Solution (see `p6-17.xls`)

- **\$10 bills (X_1):** 1, **\$5 bills (X_2):** 2, **\$1 bills (X_3):** 1, **Quarters (X_4):** 7, **Dimes (X_5):** 1, **Nickels (X_6):** 2, **Pennies (X_7):** 3
- **Minimum Total Bills and Coins Used:** 17

Spreadsheet Row	**\$10**	**\$5**	**\$1**	**\$0.25**	**\$0.10**	**\$0.05**	**\$0.01**	**LHS**	**Sign**	**RHS**
Number used	1	2	1	7	1	2	3			
Objective coeff	1	1	1	1	1	1	1	**17**	*(Objective)*	
Amount req'd	10.00	5.00	1.00	0.25	0.10	0.05	0.01	22.98	=	22.98
\$10 available	1	0	0	0	0	0	0	1	$\leq$	1
\$5 available	0	1	0	0	0	0	0	2	$\leq$	4
\$1 available	0	0	1	0	0	0	0	1	$\leq$	1
\$0.25 available	0	0	0	1	0	0	0	7	$\leq$	8
\$0.10 available	0	0	0	0	1	0	0	1	$\leq$	1
\$0.05 available	0	0	0	0	0	1	0	2	$\leq$	5
\$0.01 available	0	0	0	0	0	0	1	3	$\leq$	6

Problem 6-18: Stockbroker Susan Shader's Portfolio

Stockbroker Susan Shader has made the following recommendations to her client:

Investment	Cost (in thousands)	Expected Return (in thousands)
Andover municipal bonds	\$500	\$50
Hamilton city bonds	\$1,000	\$100
East Power & Light Co.	\$350	\$30
Nebraska Electric Service	\$490	\$45
Southern Gas and Electric	\$700	\$65
Manuel Products Co.	\$270	\$20
Builders Paint Co.	\$800	\$90
Rest Easy Hotels Co.	\$400	\$35

The client agrees to this list but provides several conditions: (1) No more than \$3,000,000 can be invested, (2) the money is to be spread among at least five investments, (3) no more than one type of bond can be purchased, (4) at least two utility stocks must be purchased, and (5) at least two regular stocks must be purchased. Formulate this as a 0–1 IP problem for Ms. Shader to maximize expected return. Solve using Excel.

Solution to Problem 6-18

1. Decision Variables

Let each binary decision variable equal 1 if the investment is chosen and 0 otherwise:

- X_1 = Andover municipal bonds, X_2 = Hamilton city bonds *(Bonds)*
- X_3 = East Power & Light Co., X_4 = Nebraska Electric Service, X_5 = Southern Gas and Electric *(Utility stocks)*
- X_6 = Manuel Products Co., X_7 = Builders Paint Co., X_8 = Rest Easy Hotels Co. *(Regular stocks)*

2. 0–1 Integer Programming Formulation

Maximize Expected Return (in \$1,000s) $= 50X_1 + 100X_2 + 30X_3 + 45X_4 + 65X_5 + 20X_6 + 90X_7 + 35X_8$

subject to:

$$
\begin{aligned}
500X_1 + 1{,}000X_2 + 350X_3 + 490X_4 + 700X_5 + 270X_6 + 800X_7 + 400X_8 &\le 3{,}000 && \text{(Budget)} \\
X_1 + X_2 &\le 1 && \text{(Max 1 bond)} \\
X_3 + X_4 + X_5 &\ge 2 && \text{(Min 2 utility)} \\
X_6 + X_7 + X_8 &\ge 2 && \text{(Min 2 regular)} \\
X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 + X_8 &\ge 5 && \text{(Min 5 total)} \\
X_i &\in \{0, 1\} && \text{for } i = 1, \ldots, 8
\end{aligned}
$$

3. Optimal Solution (see `P6-18.xls`)

- **Selected ($X_i = 1$):** Hamilton city bonds (X_2), East Power & Light (X_3), Nebraska Electric Service (X_4), Manuel Products (X_6), and Builders Paint (X_7)

- **Not Selected ($X_i = 0$):** Andover municipal bonds (X_1), Southern Gas and Electric (X_5), and Rest Easy Hotels (X_8)
- **Maximum Expected Return:** \$285,000 (\$285 thousand, at a total cost of \$2,910,000)

Row	**Andover**	**Hamilton**	**East**	**Nebraska**	**Southern**	**Manuel**	**Builders**	**Rest**	**LHS**	**Sign**	**RHS**
Invest? (1=Yes)	0	1	1	1	0	1	1	0			
Return (\$'000)	50	100	30	45	65	20	90	35	**\$285**	*(Objective)*	
Cost (\$'000)	500	1,000	350	490	700	270	800	400	2,910	$\leq$	3,000
Max 1 bond	1	1	0	0	0	0	0	0	1	$\leq$	1
Min 2 utility	0	0	1	1	1	0	0	0	2	$\geq$	2
Min 2 regular	0	0	0	0	0	1	1	1	2	$\geq$	2
Min 5 total	1	1	1	1	1	1	1	1	5	$\geq$	5

Problem 6-19: Peters Financial Investment Portfolio

Peters Financial needs to develop an investment portfolio for Mrs. Charles from the following list of possible investments:

Investment	Cost	Expected Return
A	\$5,000	\$500
B	\$8,000	\$640
C	\$3,500	\$390
D	\$10,000	\$700
E	\$8,500	\$750
F	\$12,000	\$1,000
G	\$4,000	\$300

Mrs. Charles has a total of \$30,000 to invest. The following conditions must be met: (1) If investment F is chosen, then investment G must also be part of the portfolio, (2) at least four investments should be chosen, and (3) of investment A and B, exactly one must be included. What stocks should be included in Mrs. Charles's portfolio?

Solution to Problem 6-19

1. Decision Variables

Let $A, B, C, D, E, F, G \in \{0, 1\}$ be binary variables where a value of 1 indicates that the corresponding investment is included in Mrs. Charles's portfolio, and 0 indicates that it is not.

2. 0–1 Integer Programming Formulation

Maximize Expected Return $= \$500A + \$640B + \$390C + \$700D + \$750E + \$1{,}000F + \$300G$

subject to:

$$
\begin{aligned}
5A + 8B + 3.5C + 10D + 8.5E + 12F + 4G &\leq 30 && \text{(Budget available, in \$1,000s)} \\
F &\leq G && \text{(If F is chosen, G must be chosen: } F - G \leq 0\text{)} \\
A + B + C + D + E + F + G &\geq 4 && \text{(At least 4 investments chosen)} \\
A + B &= 1 && \text{(Exactly one of A and B included)} \\
A,\ B,\ C,\ D,\ E,\ F,\ G &\in \{0, 1\} && \text{(Binary decisions)}
\end{aligned}
$$

3. Optimal Solution (see `P6-19.xls`)

- **Selected Investments:** A ($A = 1$), E ($E = 1$), F ($F = 1$), and G ($G = 1$)
- **Not Selected:** B ($B = 0$), C ($C = 0$), and D ($D = 0$)
- **Maximum Expected Return:** \$2,550.00 (using \$29,500 of the \$30,000 budget)

Spreadsheet Row	A	B	C	D	E	F	G	LHS	Sign	RHS
Selected? (1 = yes)	1	0	0	0	1	1	1			
Return (\$)	500	640	390	700	750	1,000	300	**\$2,550**	*(Objective)*	
\$ available (\$'000)	5.0	8.0	3.5	10.0	8.5	12.0	4.0	29.5	$\leq$	30.0
F and G ($F \leq G$)	0	0	0	0	0	1	0	1	$\leq$	1 ($= G$)
At least 4	1	1	1	1	1	1	1	4	$\geq$	4
A and B	1	1	0	0	0	0	0	1	$=$	1

Problem 6-21: Osaka Bombers Free Agent Pitchers

The Osaka Bombers have \$13 million available to sign free agent pitchers for the next season. The following table provides the relevant information for eight pitchers who are available for signing, such as whether each throws right or left handed, whether each is a starter or reliever, the cost in millions of dollars to sign each, and the relative value of each on the market on a scale of 1 to 10 (10 = highest):

Pitcher	Throws	Start/Relief	Cost (in millions)	Value
A	R	S	\$2	4
B	R	S	\$4	5
C	R	S	\$5	6
D	R	R	\$3	5
E	R	R	\$6	8
F	L	S	\$5	5
G	L	S	\$8	7
H	L	R	\$9	8

The Bombers feel the following needs exist for next season: (1) at least two right-handed pitchers, (2) at least one left-handed pitcher, (3) at least two starters, and (4) at least one right-handed reliever. Who should the Bombers try to sign with the objective to maximize total value?

Solution to Problem 6-21

1. Decision Variables

Let $A, B, C, D, E, F, G, H \in \{0, 1\}$ be binary variables where 1 indicates that the pitcher is signed and 0 indicates that the pitcher is not signed.

2. 0–1 Integer Programming Formulation

$$\text{Maximize Total Value} = 4A + 5B + 6C + 5D + 8E + 5F + 7G + 8H$$

subject to:

$$\begin{aligned}
2A + 4B + 5C + 3D + 6E + 5F + 8G + 9H &\leq 13 && \text{(Budget available, in \$ millions)} \\
A + B + C + D + E &\geq 2 && \text{(At least 2 right-handed pitchers)} \\
F + G + H &\geq 1 && \text{(At least 1 left-handed pitcher)} \\
A + B + C + F + G &\geq 2 && \text{(At least 2 starting pitchers)} \\
D + E &\geq 1 && \text{(At least 1 right-handed reliever)} \\
A,\ B,\ C,\ D,\ E,\ F,\ G,\ H &\in \{0, 1\} && \text{(Binary decisions)}
\end{aligned}$$

3. Optimal Solution (see `P6-21.xls`)

- **Pitchers Signed:** Pitcher A ($A = 1$, RHP Starter, \$2M), Pitcher E ($E = 1$, RHP Reliever, \$6M), and Pitcher F ($F = 1$, LHP Starter, \$5M)
- **Pitchers Not Signed:** B, C, D, G, H ($= 0$)
- **Maximum Total Value:** 17 (spending the entire \$13 million budget)

Spreadsheet Row	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**LHS**	**Sign**	**RHS**
Selected? (1 = yes)	1	0	0	0	1	1	0	0			
Value	4	5	6	5	8	5	7	8	**17**	*(Objective)*	
Budget ($Million)	2	4	5	3	6	5	8	9	13	$\leq$	13
At least 2 RH	1	1	1	1	1	0	0	0	2	$\geq$	2
At least 1 LH	0	0	0	0	0	1	1	1	1	$\geq$	1
At least 2 ST	1	1	1	0	0	1	1	0	2	$\geq$	2
At least 1 RHR	0	0	0	1	1	0	0	0	1	$\geq$	1

Problem 6-22: Allied Products R&D Project Selection

Allied Products has six R&D projects that are potential candidates for selection during the upcoming fiscal year. The table below provides the expected net present value (NPV) and capital requirements over the next five years for each project.

		Capital Required (in thousands)				
Project	**NPV (in thousands)**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
1	\$182	\$80	\$25	\$22	\$18	\$10
2	\$167	\$95	\$40	\$5	\$10	\$35
3	\$114	\$58	\$17	\$14	\$12	\$12
4	\$98	\$32	\$24	\$10	\$6	\$7
5	\$250	\$115	\$25	\$25	\$10	\$0
6	\$130	\$48	\$20	\$12	\$32	\$40
R&D Budget		\$225	\$80	\$60	\$50	\$50

The table also indicates the planned budget expenditures for the entire R&D program during each of the next five years. Which projects should be selected?

Solution to Problem 6-22

1. Decision Variables

Let $P_i \in \{0, 1\}$ for $i = 1, 2, 3, 4, 5, 6$, where $P_i = 1$ if Project i is selected and $P_i = 0$ otherwise.

2. 0–1 Integer Programming Formulation

$$\text{Maximize Total Expected NPV (in \$1,000s)} = 182P_1 + 167P_2 + 114P_3 + 98P_4 + 250P_5 + 130P_6$$

subject to:

$$\begin{aligned}
80P_1 + 95P_2 + 58P_3 + 32P_4 + 115P_5 + 48P_6 &\leq 225 && \text{(Year 1 budget, in \$1,000s)} \\
25P_1 + 40P_2 + 17P_3 + 24P_4 + 25P_5 + 20P_6 &\leq 80 && \text{(Year 2 budget, in \$1,000s)} \\
22P_1 + 5P_2 + 14P_3 + 10P_4 + 25P_5 + 12P_6 &\leq 60 && \text{(Year 3 budget, in \$1,000s)} \\
18P_1 + 10P_2 + 12P_3 + 6P_4 + 10P_5 + 32P_6 &\leq 50 && \text{(Year 4 budget, in \$1,000s)} \\
10P_1 + 35P_2 + 12P_3 + 7P_4 + 0P_5 + 40P_6 &\leq 50 && \text{(Year 5 budget, in \$1,000s)} \\
P_1,\ P_2,\ P_3,\ P_4,\ P_5,\ P_6 &\in \{0, 1\} && \text{(Binary decisions)}
\end{aligned}$$

3. Optimal Solution (see `P6-22.xls`)

- **Projects Selected:** Project 4 ($P_4 = 1$), Project 5 ($P_5 = 1$), and Project 6 ($P_6 = 1$)
- **Projects Not Selected:** Project 1 ($P_1 = 0$), Project 2 ($P_2 = 0$), and Project 3 ($P_3 = 0$)
- **Maximum Total NPV:** \$478,000 (\$478 thousand)

Spreadsheet Row	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**LHS**	**Sign**	**RHS**
Selected? (1 = yes)	0	0	0	1	1	1			
NPV ($'000)	182	167	114	98	250	130	**$478**	*(Objective)*	
Year 1 budget	80	95	58	32	115	48	195	$\leq$	225
Year 2 budget	25	40	17	24	25	20	69	$\leq$	80
Year 3 budget	22	5	14	10	25	12	47	$\leq$	60
Year 4 budget	18	10	12	6	10	32	48	$\leq$	50
Year 5 budget	10	35	12	7	0	40	47	$\leq$	50

Problem 6-25: Georgia Atlantic Corporation Warehouse Location

Georgia Atlantic Corporation needs to decide on the locations for two new warehouses. The candidate sites are Philadelphia, Tampa, Denver, and Chicago. The following table provides the monthly capacities and the monthly fixed costs for operating warehouses at each potential site:

Warehouse	Monthly Capacity (units)	Monthly Fixed Cost
Philadelphia	250	\$1,000
Tampa	260	\$800
Chicago	280	\$1,200
Denver	270	\$700

The warehouses will need to ship to three marketing areas: North, South, and West. Monthly requirements are 200 units for North, 180 units for South, and 120 units for West. The following table provides the cost to ship one unit between each location and destination:

Warehouse	North	South	West
Philadelphia	\$4	\$7	\$9
Tampa	\$6	\$3	\$11
Chicago	\$5	\$6	\$5
Denver	\$8	\$10	\$2

In addition, the following conditions must be met by the final decision: (1) A warehouse must be opened in either Philadelphia or Denver, and (2) if a warehouse is opened in Tampa, then one must also be opened in Chicago. Which two sites should be selected for the new warehouses to minimize total fixed and shipping costs?

Solution to Problem 6-25

1. Decision Variables

- **Shipping Variables:** Let PN, PS, PW be the monthly units shipped from Philadelphia to North, South, and West; TN, TS, TW from Tampa; CN, CS, CW from Chicago; and DN, DS, DW from Denver.
- **Binary Location Variables:** Let $P, T, C, D \in \{0, 1\}$ equal 1 if a warehouse is opened in Philadelphia, Tampa, Chicago, or Denver, respectively, and 0 otherwise.

2. Mixed-Integer Programming Formulation

$$\begin{aligned}\text{Minimize Total Cost} = {} & 4PN + 7PS + 9PW + 6TN + 3TS + 11TW + 5CN + 6CS + 5CW \\ & + 8DN + 10DS + 2DW + 1{,}000P + 800T + 1{,}200C + 700D\end{aligned}$$

subject to:

$$
\begin{aligned}
PN + PS + PW &\leq 250P && \text{(Philadelphia capacity)}\\
TN + TS + TW &\leq 260T && \text{(Tampa capacity)}\\
CN + CS + CW &\leq 280C && \text{(Chicago capacity)}\\
DN + DS + DW &\leq 270D && \text{(Denver capacity)}\\
PN + TN + CN + DN &\geq 200 && \text{(North requirement)}\\
PS + TS + CS + DS &\geq 180 && \text{(South requirement)}\\
PW + TW + CW + DW &\geq 120 && \text{(West requirement)}\\
P + D &\geq 1 && \text{(Open Philadelphia or Denver)}\\
T &\leq C && \text{(If Tampa opens, Chicago opens)}\\
P + T + C + D &= 2 && \text{(Select 2 warehouses)}\\
\text{All shipping variables} &\geq 0, \quad P, T, C, D \in \{0, 1\}
\end{aligned}
$$

3. Optimal Solution (see `P6-25.xls`)

- **Warehouses Opened:** Philadelphia ($P = 1$) and Denver ($D = 1$); Tampa ($T = 0$) and Chicago ($C = 0$) are not opened.
- **Optimal Monthly Shipments:** Philadelphia ships $PN = 200$ units to North and $PS = 50$ units to South ($PW = 0$); Denver ships $DS = 130$ units to South and $DW = 120$ units to West ($DN = 0$). All shipments from Tampa and Chicago are 0.
- **Minimum Total Monthly Cost:** \$4,390.00 (\$1,700 fixed cost + \$2,690 shipping cost)

Spreadsheet Row	**PN**	**PS**	**DS**	**DW**	**P**	**T**	**C**	**D**	**LHS**	**Sign / RHS**
Solution values	200	50	130	120	1	0	0	1		
Unit / Fixed Cost (\$)	4	7	10	2	1,000	800	1,200	700	**\$4,390**	*(Objective)*
Phil capacity	1	1	0	0	0	0	0	0	250	≤ 250 ($= 250P$)
TB capacity	0	0	0	0	0	0	0	0	0	≤ 0 ($= 260T$)
Chi capacity	0	0	0	0	0	0	0	0	0	≤ 0 ($= 280C$)
Den capacity	0	0	1	1	0	0	0	0	250	≤ 270 ($= 270D$)
TB or Chi ($T \leq C$)	0	0	0	0	0	1	0	0	0	≤ 0 ($= C$)
North require	1	0	0	0	0	0	0	0	200	≥ 200
South require	0	1	1	0	0	0	0	0	180	≥ 180
West require	0	0	0	1	0	0	0	0	120	≥ 120
Phil or Denver	0	0	0	0	1	0	0	1	2	≥ 1
2 warehouses	0	0	0	0	1	1	1	1	2	$= 2$

Note: Columns with zero optimal shipments ($PW, TN, TS, TW, CN, CS, CW, DN = 0$) are omitted from the summary table above for readability; see `P6-25.xls` for the full 16-column layout.